EXECUTIVE SUMMARY

Discount Impact and Profitability Analytics: A Data-Driven Approach to Financial Decision-Making

1. Overview

This project presents an interactive Business Analytics dashboard designed to examine the relationship between discount strategies, revenue, and profitability. Developed using Python, Pandas, Plotly, and Streamlit, the dashboard combines data visualization with interactive scenario modelling to support informed financial decisions.

For government stakeholders, this analytical approach demonstrates how data-driven tools can improve financial monitoring, scenario evaluation, and the transparency of decision-making processes. With appropriate adaptation and validation, similar analytical frameworks could support the assessment of pricing policies, procurement costs, revenue-generating activities, and the financial implications of alternative policy scenarios.

2. Key Findings

The dashboard's displayed scenario uses an additional discount assumption of 5% and reports the following financial indicators:

Financial Indicator	Dashboard-Reported Value
Additional discount	5%
Estimated revenue	$34,945,497.59
Estimated profit	$2,579,035.39

Estimated profit margin	7.38%
Displayed change in profit	+$1,083,256.72

These figures represent the dashboard's reported outputs and have not been independently verified against the underlying dataset.

The dashboard also presents a distribution of discount impact ratios, providing a visual overview of the variation in the displayed observations.

Important validation requirement: The dashboard states that an additional discount reduces revenue while total costs remain unchanged. Under those assumptions, profit would ordinarily decrease if revenue decreases. The reported positive profit change therefore requires investigation before the results can inform policy or financial decisions.

3. Strategic Relevance for Government Stakeholders

The project demonstrates three capabilities relevant to public-sector financial management.

Evidence-based decision-making: Interactive scenario modelling allows decision-makers to examine the financial implications of alternative assumptions rather than relying exclusively on static reports.

Financial transparency: Presenting revenue, profit, and profit margin together enables stakeholders to evaluate multiple financial indicators and identify potential trade-offs.

Improved analytical oversight: Visualizing the distribution of financial indicators can help analysts identify unusual patterns, investigate outliers, and prioritize further examination.

These capabilities could be adapted to public-sector applications, including procurement analysis, expenditure monitoring, service-delivery cost assessment, and evaluation of revenue-related policy options. Such applications would require appropriate government data, validated analytical methods, and relevant institutional safeguards.

4. Recommendations

The following actions are recommended before using the dashboard to support consequential decisions:

1. **Validate the calculations:** Verify the revenue, cost, profit, and profit-margin formulas, including the reported increase in profit.
2. **Establish a baseline:** Compare scenario results with the original financial position and clearly quantify changes.
3. **Strengthen scenario analysis:** Evaluate multiple discount assumptions and identify their effects on financial performance.
4. **Improve data governance:** Document data sources, assumptions, calculation methods, and validation procedures.
5. **Support accountable decisions:** Present analytical findings alongside relevant operational, economic, and policy considerations.
6. **Pilot before implementation:** Test any adapted public-sector application using representative data and documented performance criteria before relying on it for policy decisions.

5. Conclusion

The Customer, Product and Profitability Analytics dashboard demonstrates the potential of interactive analytics to make financial information more accessible and support structured evaluation of alternative scenarios.

For government stakeholders, its principal value lies in the analytical approach: combining visual evidence, financial indicators, and scenario modelling within a single decision-support interface.

The dashboard should currently be considered a demonstration of analytical capability rather than a validated government financial assessment. Verification of its calculations, transparent documentation of its assumptions, and testing against reliable data are necessary before operational adoption.

Overall recommendation: Support the continued development of data-driven financial analysis tools, subject to appropriate validation, data governance, transparency, and accountability requirements.